River Water Cleaning & Rejuvenation (WTE)

Large-Scale Surface Water Reclamation and Ecological Restoration.

Our River Water Cleaning and Rejuvenation (WTE) systems combine high-capacity sand filtration, biological riverbed aeration, and natural floating wetlands to restore polluted surface water bodies. Designed for municipal canals, lakes, and rivers, this ecological engineering approach removes heavy organic load, odors, and algae, returning the water body to a thriving ecological state.

Key Features & Applications

KEY HIGHLIGHTS: **- Combines mechanical filtration with biological wetlands** **- Oxygenates water to eliminate anaerobic odors and sludge** **- Tailored for high flow rates in rivers, lakes, and channels** **- Improves biodiversity and local community water access**	**COMMON APPLICATIONS:** **- Urban canal restoration projects** **- Municipal lake cleanup and aeration** **- Industrial estate run-off channels** **- Community river rehabilitation sites**

System Process & Flow

Stage / Phase	Description
Intake Clarification	**Coagulants and lamella separators filter mud, silt, and sand out of the river flow.**
Bio-Aeration	**Micro-bubbler arrays oxygenate the riverbed, breaking down sludge deposits and ending odors.**
Wetland Polishing	**Floating wetlands absorb nitrates, phosphates, and ammonia through natural root uptake.**